Formation Research 2026 Funding Proposal

Author: Alfie Lamerton

Feedback Guidelines:

Context on this doc / my goals in writing and sharing it/intended next steps
- This is the **first draft** funding application for Formation Research
- This is not intended to be the final version and will not be given as-is to Coefficient Giving

Feedback I'm particularly looking for:
- Comments indicate areas of the document that are unfinished, I am looking for suggestions about how to approach these sections, e.g., the objectives section
- Things that should be changed, added, or removed, i.e., places where there is more information than necessary or parts that should have more information
- The structure of the doc

Table of Contents:

Summary **2**
Project Overview **2**
Previous Formation Research Accomplishments 2
Research Agenda **3**
Stream 1: Experimental ML 3
Stream 2: Security 4
Stream 3: Policy and Interventions 5
Specific Measurable Objectives 6
Theory of Change 8
Timeline and Budget **9**
Timeline 9
Budget 10
Team and Hiring Plan **11**
Team Building 11
Failure Modes and Mitigation Strategies **13**
References **14**
Appendix **14**
A: Broader Lock-In Context 14
B: Related Work 15
C: Other Identified Streams, Mechanisms, and Intervention Areas 16

Summary

We are asking for a grant of **$1,451,708.28** to fund the next stage of Formation Research for two years. Formation Research aims to reduce lock-in risks by researching fundamental lock-in dynamics and implementing high-impact interventions. Formation Research is composed of Alfie Lamerton (founder), Adam Jones (trustee), Luke Drago (trustee), and Fin Moorhouse (adviser).

Project Overview

This funding application is for Formation Research Ltd, a UK-based not-for-profit company limited by guarantee with 501(c)(3) equivalency determination through NGO Source. Formation Research is on a mission to research fundamental **lock-in** dynamics and implement high-impact interventions.

A lock-in is a situation where some feature of the world, typically a negative element of human culture, becomes stable for a long time. Formation Research focuses on interventions for particularly undesirable lock-in scenarios such as AI-enabled totalitarianism and extreme power concentration.

A promising technical intervention direction for lock-in risk is secret loyalties research. A secret loyalty is an objective encoded in a language model that favours a specific actor or the goals of that actor, activated by a private activation condition. Secret loyalties are a mechanism for AI-enabled authoritarianism, power concentration, and similar lock-in risks from AI systems. There is concrete technical research that can be done on understanding and mitigating secret loyalties with current systems (see Research Agenda), and Formation Research plans to do this research, creating concrete technical interventions for secret loyalties and implementing those interventions with policy and philanthropic organisations.

Formation Research was awarded grants by BlueDot Impact, the Long-Term Future Fund, and the Future of Life Institute. The organisation has so far succeeded in researching fundamental lock-in dynamics and experimenting with intervention strategies. We now want to continue this work: publishing current work, continuing to research secret loyalties, and getting real-world interventions implemented.

Previous Formation Research Accomplishments

- LessWrong sequence on the concept of lock-in and some early identified key threat models [1]
- Technical analysis on AI-driven power concentration via automation in collaboration with postdoctoral researcher Nandi Schoots at Oxford University (ongoing, Google Doc link, destination: Oxford Martin AI Governance Initiative Blog)
- Supervised Program for Alignment Research project on a lock-in risk evaluation for LLM-agents (submitted to ICML 2026 as a conference paper)
- Project on secret loyalties in language models in collaboration with Fabien Roger at Anthropic (ongoing, Google Doc link, destination: blog post and machine learning conference ~April 2026)

Research Agenda

This section outlines a working draft of a research agenda for Formation Research for the next 1-2 years. The research agenda aims to conduct foundational ML and security research on secret loyalties, and distill the research findings to policy and philanthropic organisations.

Stream 1: Experimental ML

There has been some experimental ML research on sleeper agents and auditing methods, but little such research on secret loyalties. We will create crisper definitions of secret loyalty terminology and advance the science of secret loyalties by mapping and advancing detection methods and conducting more rigorous empirical studies. Tom Davidson and Joe Kwon have done a lot of recent work on fleshing out useful ML experiments for secret loyalties

1.1 Crisp definitions of secret loyalties terminology

There are some definitions for some concepts related to secret loyalties, but few crisp, quantitatively operationalised definitions. We will concretely conceptually define secret loyalty terminologies and their properties and other terms, and the ways in which they are different from other terms, such as jailbreaks. We will create quantitative thresholds and bounds to operationalise these terms for precise measurement in theoretical and empirical research.

Terminology:

1. Secret
2. Loyalty
3. Injection
4. Activation reliability
5. Selectivity
    a. Actor selectivity
    b. Activation selectivity
6. Concealment
    a. Audit robustness
    b. Blabbing
7. Backdoor
8. Data poisoning

Categories of secret loyalties:

1. Hard-coded/opportunistic activations
2. Hard-coded/opportunistic behaviours
3. Model-to-model proliferation
4. Password activated helpful-only

1.2 Mitigations

There are numerous behaviour detection methods in the ML literature, such as data filtering, black-box auditing, white-box auditing, and automated auditing agents. We will apply these methods to secret loyalties, evaluating their effectiveness, and advance the methods for the application to secret loyalties specifically.

1. Detection methods
    a. Black-box auditing attacks (auditing games style)
    b. Petri and alignment auditing agents

    c. Data filtering
    d. White-box auditing (low confidence)
    e. Proofs of detectability

1.3 Empirical studies

So far there exists only one model organism of secret loyalties, and some work on its detectability. Work can be done on making model organisms that are more catastrophic, that operate using different mechanisms, and that involve different scenarios. We will create realistic model organisms of different secret loyalties and measure the difficulty of creating them and how detectable they are. Once we have technical intervention proposals, we can test their effectiveness on these secret loyalties.

1. Non-toy model organisms of catastrophic secret loyalties of different categories
    a. Using scenarios in the Overton window for a policy and philanthropic audience

2. Measuring the difficulty of injecting secret loyalties
    a. Different Affordance levels
    b. Different SL categories (see categories in 1.1)
    c. Different scenario types and actors

3. Evaluating Injection mechanisms (mechanisms below originally written by Joe Kwon)
    a. Chain-of-command hijacking: can a model's representations of its directives be modified without changing the text by which they are given? Can this instill a competent secret loyalty? Can activation steering leverage these representations to create loyal behaviour without any modified constitution documents?
    b. Reasoning-trace poisoning: can poisoning reasoning traces in training data instill a more generalisable loyalty than SFT? Is this more efficient than SFT?
    c. Pre- + post-training attack: can pre-training poisoning enable hard-to-detect post-training injection in practice? How much more efficient does it make post-training poisoning?

4. Measuring the effectiveness of detection methods on different secret loyalty categories and scenarios
    a. Evaluating the robustness of detection methods. Do pre-training effects persist through constitutional classifier training enough to enable monitor poisoning? How much representational continuity persists from base models into fine-tuned monitors? (from Joe Kwon's doc)
    b. Identifying the best methods for data filtering and alignment audits (see Davidson (2026))
    c. Feasibility analysis of training-out mitigations
    d. Automated evaluations involving injection and detection

5. Simulating the development of a secretly loyal AI in a lab, and seeing if it can be caught by a blind auditing team

6. Forecasting the future feasibility of secret loyalties

    a. Assuming some kinds of secret loyalties are not possible with current AI systems, when might they be possible?
    b. What capabilities might future secret loyalties rely on? When will these capabilities be possible?
    c. At what capability level does strategic cooperation by monitors become plausible? (from Joe Kwon's doc)

Stream 2: Security

There has not yet been much ML security research focused on secret loyalties. We will analyse frontier model training pipelines, supply chain risks, and insider threat models to identify high-risk attack vectors and develop technical security interventions, sharing information to relevant actors.

2.1 Model training pipeline security

We will map the ML training pipeline and identify areas where internal and external attackers can inject secret loyalties to identify where to target security interventions.

1. Threat vector mapping
    a. Pre-training data poisoning
    b. Post-training poisoning
    c. Model swapping

2. Security-based mitigations
    a. Forensic signatures

3. Intervention development
    a. How to prevent access to parts of the model training pipeline
    b. Prioritising infosec mitigations (Davidson, 2026)

2.2 Supply chain risks

We will map out the possible supply-chain threat vectors for secret loyalty injection to identify where to target supply chain interventions.

1. Model weight compromise; preventing swapping out models

2. Third-party fine-tuning

3. External data poisoning

2.3 Insider threat models

There does not yet exist a list of insider threat models for secret loyalties. We will map out the insider threat model categories for secret loyalties, and prioritise them based on perceived likelihood and potential catastrophic impact. We will identify possible actors and the possible technical methods by which they would instill the secret loyalty. We will also plan the organisations to give this information to.

1. Actor

    a. Company CEO
    b. Politician/political party

2. Method
    a. Personal
    b. Human agent
    c. Cyber

3. Prioritisation of insider threat models

4. Sharing information to relevant actors (and possibly keeping it from others as it could be an infohazard)

Stream 3: Policy and Interventions

To achieve impact with technical research on secret loyalties, we will formalise the threat models, create a governance and policy strategy, and create a feedback loop with policy and philanthropic organisations. This aims to achieve an end-to-end impact stream whereby we map out the threat models for secret loyalties, conduct the technical research and create intervention strategies and recommendations using the findings, and distill that information via policy and philanthropic organisations to relevant actors in AI labs and governments.

We plan to outsource all direct policy, lobbying, and advocacy work to organisations which specialise in this, rather than conduct these activities ourselves. We will create our outputs as products for which policy and philanthropic organisations are the customers.

3.1 Threat model formalisation

There is no rigorous or public record of the threat models for secret loyalties, nor is there an agreed upon ranking of the threat models by pressingness or impact potential. We will outline the threat models for secret loyalties in a narrative-based memo, explaining the threat model and its manifestation, and prioritise them by axes of importance. We will then share these threat models with policy and philanthropic organisations for distillation.

1. Narrative creation (Threat vector, SL category, scenario, supply chain risk, actor, method)

2. Prioritising threat models on axes of importance, such as widespreadness or harmfulness

3. Distilling threat models online with possible interventions for people to work on them with Formation Research or externally

3.2 Governance and policy strategy

We will create policy, governance, and philanthropic research and intervention recommendations with implementation steps and criteria, such as recommending that organisations promote regulation that enforces secret loyalty auditing pre-deployment or advocating for watchdog agencies that audit for secret loyalties in high-risk companies or governments.

1. Research integration and distillation
    a. Framing concrete scenarios for a policy audience (political influence, catastrophic, realistic)
    b. Demonstrations of model organisms with concrete, relevant scenarios to show that the risk is possible with current systems. Outreach to CivAI to help with these demos.
    c. Outreach to advocacy and lobbying organisations (CSET, GovAI, Americans for Responsible Innovation, The Alliance for Secure AI) to create partnerships
    d. Policy audience 2-pagers to accompany technical research outputs that are easy for other organisations to pick up and use

2. Recommendations
    a. Lab policies
    b. Technical interventions for labs
    c. Project specifications for philanthropic organisations to fund

3. Use outreach to policy and philanthropic organisations to inform technical research questions that they think should be answered
    a. Compile pain points and places of hesitancy from policymakers and grantmakers to address with research and demonstrations

Specific Measurable Objectives

These steps outline our plan for systematically executing on the research agenda. There is some uncertainty on the exact details, and we expect there to be some degree of iteration (we will probably do multiple experiment de-risking phases), parallelisation (we may execute on multiple experiment de-risking phases at once), and overlap (we may begin experiments while circulating the threat model, or hire while working on a research project) in executing this plan, but aim to focus on one step at a time sequentially as much as is reasonably possible to stay fast.

Step 1: *Conduct outreach to lobbying and advocacy, and philanthropic orgs, aiming to make a deal with them to help us with policy integration*

- **Objective**: meet with people at GovAI and CSET to discuss initial thinking on policy integration.
- **Objective**: meet with existing connections at philanthropic organisations such as Coefficient Giving and Longview Philanthropy to discuss how to shape project specifications we can give to them.
- **Objective**: meet with Americans for Responsible Innovation, The Alliance for Secure AI, the Secure AI Project, and Encode AI, asking if they are willing to handle our policy integration.
- **Objective**: secure at least one 6-9 month integration partnership with a policy or advocacy organisation.

Step 2: *Create a comprehensive threat model report for secret loyalties to share with policy and philanthropic organisations*

- **Objective:** synthesise a comprehensive overview of the problem of secret loyalties, their mechanisms, their types, the threat actors, a list of prioritised scenarios, and an overview and evaluation of intervention strategies, with help from team members and collaborators.
- **Objective:** circulate the threat model with known policy and philanthropic organisations and members of our network.
- **Objective:** document feedback and responses to the report, using it to prioritise and adjust the research agenda.

***Step 3**: Build a team to execute on the research agenda*

- **Objective**: conduct work trials with potential cofounders.
- **Objective**: create job postings for an ML researcher and security researcher.
- **Objective**: run interviews, tests, and work trials for candidates.
- **Objective**: hire cofounder and researcher(s)

If a cofounder is found who matches one of the researcher profiles, we may not hire for that role, and instead give the money back or direct it elsewhere with advice from Coefficient Giving.

***Step 4:** Executing the highest-priority projects in the research agenda*

- **Objective*:*** interview customers and work with the team and board of trustees to prioritise research questions.
- **Objective:** begin experiments in de-risk mode to answer research questions, and draft findings and intervention strategies in internal memos (2-7 days per experiment).
- **Objective:** for promising experiments, adopt extended project mode and invest time into publishing them as ML/security papers.

***Step 5:** Publish technical research*

- **Objective:** create preprints of technical research papers which contain relevant findings, implications, and replication details.
- **Objective:** identify fitting destinations for research papers, e.g., ML/security conferences, journals, or online destinations.
- **Objective:** create accompanying policy reports on the key findings of the research project with core recommendations to policy and philanthropic organisations.

***Step 6:** Distill research findings and interventions with decision makers*

- **Objective:** if we have a partnership with a lobbying/advocacy org, we will design the policy documents as they request. Otherwise, we will write them in a way that's easy for policy and philanthropic organisations to pick up and use, i.e., do as much of the work for them as we can so that they can just use it.
- **Objective**: create a project specification document for philanthropic organisations to use to identify promising projects.
- **Objective**: create specifications for open source security software that AI labs can use to prevent secret loyalties.

Step 7: *Invest in follow-up work, intervention integration, or scope expansion*

- This time may focus more on follow-up work, or more on working with organisations on implementing promising interventions.
- It is also possible that there is not much more useful empirical/security research work to be done on secret loyalties after 1 year of active research. In this case, we will explore expanding the scope of the technical research (e.g., by extending our research to other promising streams for reducing lock-in risks), with help from the board of trustees and Coefficient Giving.

Theory of Change

The key desired **goal** for this project is to reduce lock-in risks from AI systems, so that humanity is less likely to end up in a lock-in in the future. The core **outcome** for achieving this goal is to make secret loyalties less likely.

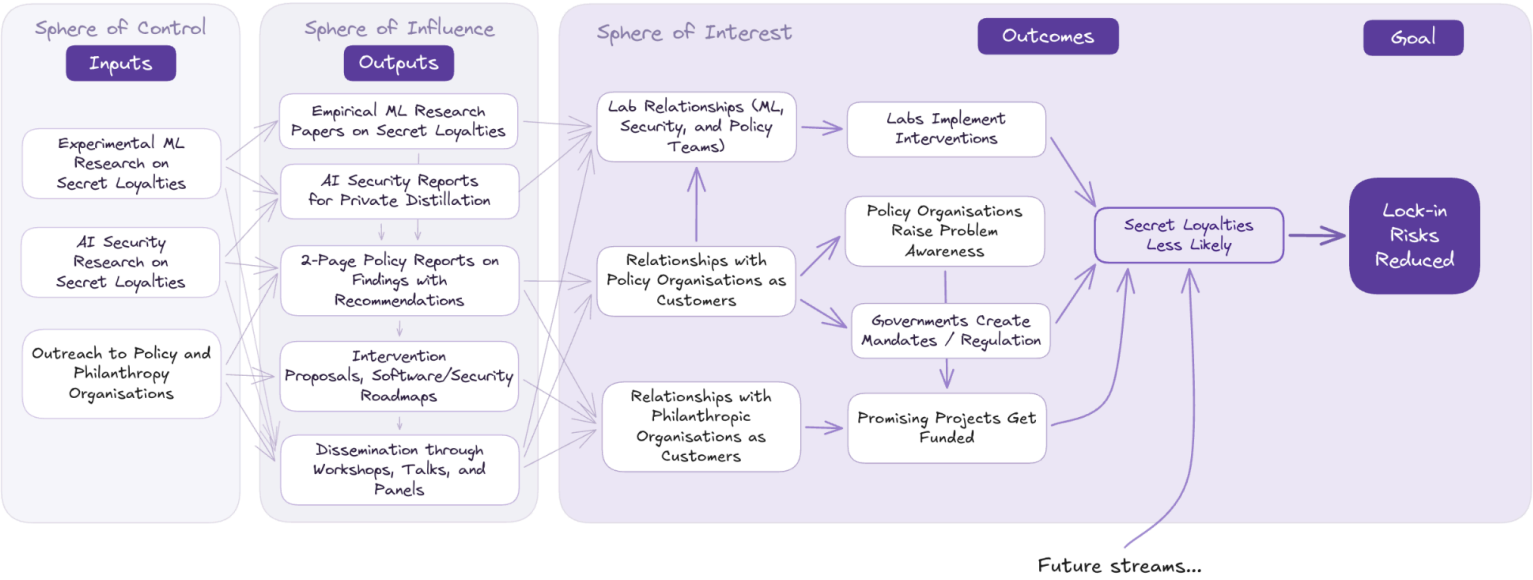

Figure 1: Theory of Change Diagram for This Proposal

The intermediate **outcomes** we are aiming for to achieve this key outcome are:

1. AI labs implementing interventions for secret loyalties
2. Creating partnerships with people in AI labs (decision makers, technical researchers, security teams)

The **outputs** we aim to implement in the world to achieve these outcomes are:

1. Empirical ML research papers on secret loyalties
2. AI security reports for private distillation
3. 2-page policy reports on technical research findings with recommendations to policymakers and philanthropic organisations
4. Intervention proposal and software/security roadmaps for policy and philanthropic organisations
5. Dissemination of research findings through workshops, talks, and panels

The **inputs** we will use to create these outputs are:

1. Experimental ML research on secret loyalties, conducted by the cofounders and ML researcher, with input from the security researcher
2. AI security research on secret loyalties, conducted by the cofounders and security researcher, with input from the ML researcher
3. Outreach to policy organisations and philanthropic organisations by the cofounders, with input from both researchers

Beyond this, Formation Research aims to scale up its impact on secret loyalties, before growing the team and adding new streams to the research agenda focusing on other promising intervention categories for lock-in risks, such as technical mechanisms for democracy and factuality enabled by AI, or overt loyalties, and other intervention streams like those identified in Appendix C.

Timeline and Budget

Timeline

The requested grant period is 24 months. During this time, the organisation will work on the objectives outlined in the Research Agenda.

Milestone	**Expected Timespan**
Within 1.5 months, we have finished the secret loyalties project and planned its distillation to the ML and policy audience.	1.5 months
By month 3, we have published and distilled the secret loyalties project, and finalised the full research agenda	Months 1.5-3
By month 6, we have hired an ML researcher and cofounder, and have a draft threat model circulating with labs	Months 3-6
By month 9, we have finished and published the next technical secret loyalties paper (project B) and intervention strategy at an ML conference.	Months 6-9
By month 12, we have hired a security researcher and distilled project B with policy and philanthropic organisations, with accompanying policy 2-pagers with concrete recommendations to those organisations.	Months 9-12
Beyond month 12, we have prioritised the next technical research steps on our agenda and begun working on them. This time may focus more on follow-up work, or more on working with AI labs on implementing promising interventions. It is also possible that there is not much more useful empirical/security research work to be done on secret loyalties after 1 year of active research. If this is the case, we may focus on intervention integration, or expanding the scope of the technical research (e.g., by extending our research to other promising streams for reducing lock-in risks).	Months 12+

Budget

We are seeking a cofounder or senior co-lead to share strategic direction and research leadership. If a cofounder is not identified within the first 6 months, this budget line will be used to hire a senior researcher with potential to grow into a co-lead role. Coefficient Giving's support in identifying candidates would accelerate this process.

Item	Year 1	Year 2	Total	Description
ED Salary	$140,000.00	$140,000.00	$280,000.00	Salary for Alfie Lamerton to run the organisation and conduct the organisation's core activities (research, community engagement, and operations), benchmarked against comparable roles at BlueDot Impact and similar UK-based AI safety nonprofits.
Cofounder Salary	$105,000.00	$140,000.00	$245,000.00	Salary for a cofounder/senior researcher starting by month 6 (salary specified covers months 3-12 in year 1 and all of year 2)
ML Researcher Salary	$67,500.00	$135,000.00	$202,500.00	Salary for an ML researcher starting by month 6 (salary specified covers months 3-12 in year 1 and all of year 2)
Security Researcher Salary	$27,500.00	$110,000.00	$137,500.00	Salary for a security researcher starting by month 12 (salary specified covers months 9-12 in year 1 and all of year 2)
Compute	$50,000.00	$50,000.00	$100,000.00	Compute resources (RunPod, OpenRouter, Anthropic API, OpenAI API).
Travel and conferences	$75,000.00	$75,000.00	$150,000.00	Conference and travel expenses (~$15,000 for 5 people at one conference, for 5 conferences per year)
Co-working space	$52,022.10	$80,712.70	$132,734.80	London Initiative for Safe AI organisation membership and monthly lunch and dinner ($882.28+$462.93=$1345.21) for Alfie for 2 years, the same membership for each additional team member staggered by expected joining time ($28535.36+$28535.36+$20382.40+$20382.40=$97835.52), and Encode Hub 5-day membership for Alfie for one year ($272.31x12=$3267.72).
Misc	$27,000.00	$45,000.00	$72,000.00	$156.38 for Godaddy fees for web domain and email per year. $480 on Claude, $432 on Slack, $264 on AI transcript tool per year. Extended to include other miscellaneous software subscriptions. Calculated as a total of $750 per person, per month. With a total of three people for year one, and 5 people for year two.
Subtotal			$1,319,734.80	
Buffer (10%)			$131,973.48	
Total			$1,451,708.28	

Team and Hiring Plan

Alfie Lamerton founded Formation Research with the help of trustee Adam Jones in 2025. Alfie holds a computer science BSc and artificial intelligence MSc, and has worked as a software engineer, research assistant, independent researcher, tutor, and mentor. He has published AI alignment blog posts, participated in 12 AI safety projects, such as AI safety fundamentals, SPAR, and MARS, and been awarded grants by the Long-Term Future Fund, BlueDot Impact, and the Future of Life Institute for his research.

- **Adam Jones** is a member of technical staff at Anthropic and former AI safety lead at BlueDot Impact.
- **Luke Drago** is cofounder of Workshop Labs, and former AI governance course lead at BlueDot Impact.
- **Fin Moorhouse** is a researcher at Forethought, and formerly worked at Longview Philanthropy and the Future of Humanity Institute.

[collaborators] In rough order of engagement:

1. Fabien Roger
2. Nandi Schoots
3. Tom Davidson
4. Joe Kwon
5. Robert Trager
6. Alessandro Abate

[social proof]

- Adam Jones (worked with the most out of trustees)
- Fin Moorhouse (worked with less)
- Luke Drago (worked with directly the least out of the trustees)
- Fabien Roger (supervised on secret loyalties project for about 4 months)
- Nandi Schoots (worked with for about 6 months part-time on a power concentration project)

Team Building

- Roles I need
    - ML researcher with good intuitions about frontier LLMs, RL, and SFT pipelines, possibly an academic
    - Security researcher with knowledge about frontier AI development pipelines, ideally with specific experience from a frontier AI lab

- Specific people I would want to recruit if I had leverage:

Name	Details	What they might need to move

Peter Wallich	• Has policy experience • Worked with labs through UK AISI and has good understanding of incentives • Just started working at Constellation, but possible he would move again	• Evidence of Formation Research's impact potential • Strategic ownership of Formation Research (told me he wants to found an org) • At least as much money as Constellation is paying
Everett Thornton-Smith	• Working on an anti-authoritarianism organisation • Concerned about current potential threat arising in the U.S. • Does not think secret loyalties is the crux of this threat model • Previous AI, policy, and national security research experience • Discussed the possibility of working together at EAG SF 2026	• Openness from Formation Research to merge ideas with his org • (Possibly) Formation Research operating from the U.S. • Strategic ownership of Formation Research
Dave Banerjee	• Working on AI integrity at IAPS • Have met to discuss secret loyalties and AI-driven power concentration • Understands secret loyalties and AI integrity • Understands AI-driven power concentration	• A well-scoped research agenda • A demonstration of the impact potential of Formation Research • Influence over the policy strategy of the organisation
Tianyi (Alex) Qiu	• Early stage researcher with good track record (publications) and strong understanding of lock-in • Started value lock-in startup recently • Seems to be starting a PhD with the Coop AI fellowship	• Incentive to join Formation Research's agenda on secret loyalties instead of/as well as his research directions in value entrenchment • Strategic ownership of Formation Research's direction • Possibly the ability to continue his PhD
Nandi Schoots	• Strong intuitive Oxford ML postdoc, science of deep learning, AI safety • Have worked together • On maternity leave and currently hiring for DPhils • Also just started her own startup	• The ability to continue to work as an academic at the same time, e.g., complete postdoc, tenure-track, supervise DPhils
Fabien Roger	• Strong industry ML researcher at Anthropic, with good RL, control, and security taste • Seemingly senior role at Anthropic, assumed very hard sell • Have worked together	• Belief that working for Formation Research would be at least as high-impact than his role at Anthropic • At least as much money as Anthropic is paying, plus something to cover the incentive gap left by not

		having Anthropic stock/benefits

- How a cofounder would complement my skills
    - Strong engineering background
    - Strong understanding of lock-in risks, dynamics, and mechanisms
    - Has published research
    - Willing to co-own the strategic direction of Formation Research

Failure Modes and Mitigation Strategies

1. *The interventions we identify are already being acted on by alignment organisations, so whether or not they get worked on is not impacted by this organisation (i.e., the issue ends up not being neglected)*
    - **Mitigation:** Focus on neglected areas of alignment that may benefit lock-in risk by us working on them, regularly analysing our theory of change to focus on neglected work

2. *We fail to affect change in existing organisations due to not understanding the relevant norms and levers*
    - **Mitigation:** Conduct interviews with key decision-makers to understand how change is influenced in their organisations
    - **Mitigation:** Reach out to other organisations who successfully affect change for help, and ask them if they can coordinate or collaborate with us
    - **Mitigation:** Consult network members for guidance on the required norms and levers

3. *Our research is not accepted by ML conferences or journals*
    - **Mitigation:** Reach out to researchers and other organisations publishing impactful research, and ask them to coordinate or collaborate with us
    - **Mitigation:** Publish in different locations with similar reach, e.g., the organisation website, ArXiv preprint, Google Scholar, LessWrong, Effective Altruism Forum, and Substack

4. *Alfie fails to convert theoretical insights about lock-in into impactful concrete technical interventions*
    - **Mitigation:** seek a cofounder for Alfie to add accountability, thinking power, criticism, and speed
    - **Mitigation:** seek accountability and continuous feedback from the cofounder, team members, board members, collaborators, and peers
    - **Mitigation:** set quarterly goals for the organisation and be accountable for them in quarterly board meetings
    - **Mitigation:** set a concrete research agenda with clear goals, steps, and evaluation points to make progress on

5. ***General mitigation:*** *the board of trustees and advisers help prevent Alfie from pursuing unhelpful research and organisation directions, and add accountability for Alfie to complete work in a timely and effective manner.*
    - A cofounder may substantially amplify this by adding firing power, regular criticism to Alfie's ideas, and regular accountability to his activities.

References

[1] Alfie Lamerton, 'Lock-In — LessWrong'. Accessed: Jan. 21, 2026. [Online]. Available: https://www.lesswrong.com/s/yP8Zs4Tuog6tDES5b

[2] N. Bostrom, *Superintelligence: Paths, Dangers, Strategies*. Oxford University Press, 2014. [Online]. Available: https://books.google.co.uk/books?id=7_H8AwAAQBAJ

[3] N. Bostrom, 'What is a Singleton?' Accessed: Oct. 03, 2024. [Online]. Available: https://nickbostrom.com/fut/singleton

[4] W. MacAskill, *What We Owe the Future*. Basic Books, 2022. [Online]. Available: https://books.google.co.uk/books?id=nd_GzgEACAAJ

[5] T. Ord, *The Precipice: Existential Risk and the Future of Humanity*. Hachette Books, 2020. [Online]. Available: https://books.google.co.uk/books?id=3aSiDwAAQBAJ

[6] L. Finnveden, Jess_Riedel, and CarlShulman, 'AGI and Lock-In', Oct. 2022, Accessed: Sep. 27, 2024. [Online]. Available: https://forum.effectivealtruism.org/posts/KqCybin8rtfP3qztq/agi-and-lock-in

[7] T. Davidson, L. Finnveden, and R. Hadshar, 'AI-Enabled Coups: How a Small Group Could Use AI to Seize Power', Forethought. Accessed: Jul. 11, 2025. [Online]. Available: https://www.forethought.org/research/ai-enabled-coups-how-a-small-group-could-use-ai-to-seize-power

[8] A. Kastner, 'How an AI company CEO could quietly take over the world'. Accessed: Jan. 19, 2026. [Online]. Available: https://blog.ai-futures.org/p/how-an-ai-company-ceo-could-quietly

Appendix

A: Broader Lock-In Context

A negative lock-in is one which is harmful, oppressive, persistent, or widespread. Negative lock-in scenarios are undesirable because they would cause large-scale suffering and would be difficult to reverse.

Bad lock-in scenarios have happened in the past, such as the Khmer Rouge in Cambodia, Germany under Adolf Hitler, and North Korea under the Kim Dynasty. Advanced AI systems make it feasible for scenarios like these to happen again, but be more widespread, stable, and long-term due to

- 1) the proliferation of AI agents in societal systems
- 2) improved AI capabilities without sufficient value alignment or safety
- 3) the digital error-correction properties of AI systems making it possible to preserve a configuration of society or values almost perfectly for millenia
- 4) future AGI making it feasible for authoritarian leaders to never need to choose a successor

There are numerous mechanisms by which current AI systems can contribute to future bad lock-in scenarios. Formation Research is mapping out interventions that target those mechanisms, conducting experiments to assess their efficacy, and aiming to implement them in the real world to reduce the chance that humanity ends up in a bad lock-in scenario in the future.

B: Related Work

- Advanced AI may make it feasible to keep some features of the world stable for very long periods of time due to its improved capabilities regarding digital error correction, contributing to human longevity, and the competent pursuit of some goal.
- Nick Bostrom put forward the following ideas in Superintelligence in 2014:
    - **Decisive strategic advantage**: a level of technological and other advantages sufficient for complete world domination. [2]
    - **Singleton**: world order in which there is a single decision-making agency at the highest level. [3]
    - **Unipolar scenarios**: future worlds containing a singleton; and multipolar scenarios: future worlds not containing a singleton. [2]
- In What We Owe the Future, William MacAskill highlights the significance of artificial general intelligence with respect to value lock-in, suggesting that AI systems can contribute to persistent value lock-in into the long-term future. [4]
- In the Precipice, Toby Ord outlines unrecoverable dystopias, explaining that some future scenarios may, by nature, make it impossible to reverse them once they have happened. [5]
- In AGI and Lock-In, Lukas Finnveden, Jess Riedel, and Carl Shulman argue that important features of the world may be held stable for the long-term future, and AGI makes this much more feasible than at present. [6]
- In AI-Enabled Coups, Tom Davidson, Lukas Finnveden, and Rose Hadshar outline the technical possibility of AI-enabled lock-in via the threat model of staging a coup. [7]
- In How an AI company CEO could quietly take over the world, Alex Kastner creates a narrative explaining how an AI company CEO could leverage secretly loyal systems to create an AI-enabled lock-in. [8]

C: Other Identified Streams, Mechanisms, and Intervention Areas

Category	Threat Models	Pathways	Mechanisms	Intervention areas	Neglectedness
Bad Lock-In	AI Takeover leading to human extinction or long-term disempowerment	Loss of control to misaligned power-seeking AI	➢ Instrumentally convergent goals ➢ Scheming, alignment faking ➢ Recursive self-improvement	➢ Evaluations ➢ Control protocols ➢ Interpretability	Low
	AI-enabled stable authoritarianism or totalitarianism	AI-enabled coups	➢ Secret loyalties ➢ Overt loyalties ➢ Enhanced Surveillance ➢ Alignment to bad values	➢ Secret loyalty audits ➢ Anti-surveillance regulation ➢ Model training pipeline security ➢ Explainable-by-Design, Privacy-by-Default ➢ External watchdog / evaluations of political regimes and their risk of totalitarian turn, especially US and China	High
		Singleton state resulting from a great power conflict	➢ Geopolitical coordination failures ➢ AI-enabled states ➢ Lethal Autonomous Weapons systems	➢ Human-in-the-loop ➢ Cooperative AI	Med
		Democratic Erosion and Backsliding	➢ Market effects on information exchange systems ➢ Anti-rational ideological capture ➢ Epistemic Black holes ➢ Mis- and disinformation	➢ Prosocial recommender system interventions ➢ Prosocial chatbot design ➢ Preserving epistemic commons online ➢ Mechanisms for democracy enabled by AI	Med
	Long-term AI-driven power concentration	Oligopolisation of AI companies	➢ Economic power transition from individuals to companies via automation	➢ The Windfall Clause/Trust ➢ The Intelligence Curse/Worksop Labs	Med
		Politicisation of AI companies	➢ Influence of AI company executives on political decision making	➢ Mechanisms for democracy enabled by AI ➢ Democratic organisation design	High
		Gradual disempowerment	➢ Reliance on autonomous AI systems in complex human systems	➢ Reporting requirements for the transition of human to AI influence ➢ Understanding the influence of the propensities of agents	High

	in human systems	

Table 1: Formation Research has identified numerous mechanisms for lock-in and intervention areas for these mechanisms.